CSC_51053_EP, Exercise Session 1: Data Modeling

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1 Entrance exams

Admission in French engineering schools is done by exams. Each candidate has a name and an ID. Each school admits candidates through one or several tracks (or majors), e.g., "Math-Physics", "Physics-Chemistry", etc. For each track, the school publishes a number of places. To be admitted in a track, candidates must first pass a fixed set of written exams. Each exam has a name, e.g., "Maths 1", and an exam topic. For each written exam, each candidate gets a grade. Also, for each written exam of one school, the school has a coefficient (weight) to give to the exam. For each track, a fixed number of candidate will be admitted to pursue to oral exams. Concretely, those candidates are those whose weighted averages (sum of grades $\times$ respective coefficients) are the highest.

Each school also determines a set of oral exams to be taken by any candidate remaining after the written exams. Each oral exam has a weight, and each candidate gets a grade. A candidate gets an offer from a school in a track, if the candidate has applied to that track of the school, and the candidate's overall grade (the sum of the candidate's oral and written exam grades, each multiplied by the respective weight) ranks the candidate at the top (within the number of places).

1. Propose a minimal Entity-Relationship model for the admission process. Which aspects of the specification are not described in the model? Briefly explain why.
2. If we want to record the date when each candidate takes an exam, where can we put it?
3. Suppose we want to add to the model the fact that to apply to a track of a school, a candidate must pay a fee that is the same for all the tracks of the school. Modify the above model to reflect that.
4. How can we change the model so that we make sure when a candidate registers to a track, the candidate has a grade for each written exam in the track?

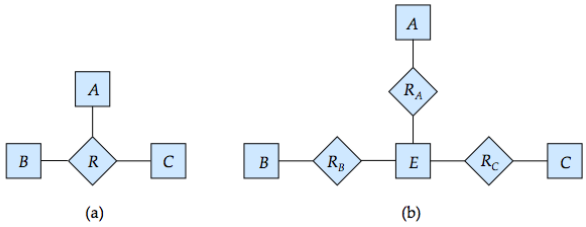
2 E-commerce database

We consider an online commerce site selling *products* (supplied by a set of *vendors*) to *clients*. Each product has an *ID*, a *name* and a *description*. Each vendor has an *ID* and an *address*. Each client has an *ID*, a *name* and an *address*. Each vendor supplies at least one product. For each product that it supplies, a vendor charges a certain *price*, and some *shipping fees*, which are determined by the address of the vendor, the address of the client, and the product itself. The *delivery time* of a product to a client is also dictated by the vendor and client addresses. The online commerce site charges clients a *margin*, which is a percentage of the price of every product they buy. This percentage is the same for all the products supplied by a given vendor but it can be different for diffent vendors.

1. **(E-R model)** Propose a set of entities and relationships that model the data described above and its constraints. If you detect constraints which cannot be expressed using the notions seen in the course, state the constraint and briefly explain why.
2. **(Relational model)** Transform this Entity-Relationship model into tables (relations) and constraints as seen in the course. Use single-table constraints whenever possible and multiple-table constraints only if necessary.

3 Exercise 2 extended from [1]: modeling ternary relationships

Consider the two alternatives below for modeling a ternary relationship R between entities of the sets A, B and C using E/R schema: The first diagram (a) is a straightforward translation of a ternary relation. The second diagram (b) is a more complex translation of R.



1. Construct a simple instance for the entity sets E, A, B, C and relationship sets R_A, R_B, R_C, which *cannot* be modeled with the entity sets A, B, C and the relationship R.

2. Propose a **diagram (b) variant (modification)** to introduce constraints that guarantee that any instance over the entities E, A, B, C and relationships R_A, R_B, R_C which satisfies the constraints, can be modeled with the entity sets A, B, C and the relationship R.

3. Now consider that in (a), entities A, B, C all have a total participation constraint to the relationship R. Modify the result of the previous point (variant on (b)) in order for the three-entities model to capture any instance of the four-entities one. Provide two variants:

   (a) Using participation constraints;

   (b) Using weak entities.

4. As a variant on this exercise, consider *four* entities A, B, C and D, and a relationship S of arity four between them. Assume we want to express the fact that *as soon as four entity instances a, b, c, d are connected through S, the entities a, b, c are also connected through R.*

   (a) Can this be done by adding entities and/or relationships to the ER model (a) at the beginning of this exercise?

   (b) Can this be done by adding entities and/or relationships to the ER model (b), modified as above in item 3a?

References

[1] H. Korth, S. Sudarshan, and A. Silberschatz. *Database System Concepts (6th ed.)*. McGraw-Hill Education, 2010.